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Invited Review Paper

Transfer learning algorithms for bearing remaining useful life prediction: A comprehensive review from an industrial application perspective

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ABSTRACT

Accurate remaining useful life (RUL) prediction for rolling bearings encounters many challenges such as complex degradation processes, varying working conditions, and insufficient run-to-failure data. Transfer learning (TL), one paradigm of artificial intelligence technology, has demonstrated its powerful performance and great effectiveness for such challenges. As a result, many TL-based solutions have been widely developed and extensively studied for rolling bearing RUL prediction. Admittedly, several review articles have been published on RUL prediction. Nevertheless, the majority of these articles only concentrated on deep learning-based RUL prediction methods, and a review article that systematically overviews the status of TL-based RUL prediction has not been published. Therefore, it is urgent and significant to thoroughly summarize the academic publications and industrial applications related to TL-based RUL prediction, and present its potential challenges and future research directions. With such goals, the problem definitions of TL-based RUL prediction, the general procedure of RUL prediction, and typical TL-based RUL prediction algorithms are first introduced to help researchers quickly overview the state-of-the-art approaches and recent developments. Thereafter, relevant TL-based RUL prediction solutions are comprehensively discussed from the perspectives of three industrial scenarios, providing suggestions to researchers and engineers for selecting appropriate solutions in practical industrial applications. Finally, the key challenges and future trends in RUL prediction are presented to conclude this paper. We hope that this review of TL-based RUL prediction for rolling bearings can contribute to a better understanding of intelligent prognostic technology and will inspire researchers to extend their work on RUL prediction.

1. Introduction

With the deep integration and rapid development of advanced artificial intelligence, sensing and information technology, prognostics and health management (PHM) [1–3], conducted by utilizing and analyzing various status signals, has become an urgent need for improving the performance of intelligent manufacturing. PHM techniques have been applied to many fields such as machine

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Nomenclature

Acronyms

Acronyms	Terminology
AE	Auto Encoder
Bi-LSTM	Bidirectional-LSTM
CNN	Convolutional Neural Network
DAAN	Dynamic Adversarial Adaptation Network
DANN	Domain-Adversarial training of Neural Networks
DBN	Deep Belief Network
EKF	Extended Kalman Filter
FL	Federated Learning
FNNs	Feedforward Neural Networks
GAN	Generative Adversarial Networks
GPR	Gaussian Process Regression
KL	Kullback Leibler
LS-SVM	Least-Square SVM
Acronyms	Terminology
LSTM	Long Short Term Memory
MMD	Maximum Mean Discrepancy
MK-MMD	Multi Kernel-MMD
PHM	Prognostics and Health Management
RNN	Recurrent Neural Network
RUL	Remaining Useful Life
SAE	Sparse Auto Encoder
SDAN	Sparse Domain Adaptation Network
SVM	Support Vector Machine
TCA	Transfer Component Analysis
TCCHC	Transfer Compact Coding for Hyperplane Classifiers
TL	Transfer Learning
WAFTR	Weibull Accelerated Failure Time Regression

manufacturing, power systems, and aerospace [4–6]. As one of the critical components of rotating machinery, rolling bearing frequently fail due to harsh working conditions, such as heavy loads and high speeds in industrial applications [7,8]. According to statistics, half of the rotary machine system faults are caused by rolling bearings [9,10]. Furthermore, bearing faults are responsible for up to 52% of failures on high-voltage induction motors in offshore oil industries [11], and 76% of failures in the majority of wind turbine gearboxes [12,13]. According to Asea Brown Boveri Ltd., the Institute of Electrical and Institute of Electrical and Electronic Engineers, and the Electric Power Research Institute, bearings are the most damaged rotating machinery in induction motors, representing 51%, 41%, and 42% of the damage, respectively [14]. Therefore, it is urgent to develop effective methods for evaluating the current health status of bearings and predicting their remaining useful life (RUL). The RUL of rolling bearings is defined as "the length from the current time to the end of the useful life" [15]. RUL prediction methods have emerged as powerful tools in PHM fields, and effective RUL prediction can provide valuable suggestions and reasonable references before making maintenance decisions, which can reduce downtime and economic losses and improve the reliability of machines.

However, different from the fault diagnosis problem, rolling bearing RUL prediction is much more challenging because of the following characteristics.

(1) RUL prediction is a regression problem, and its output value is the continuous remaining lifetime, while the fault diagnosis problem emphasizes the classification of fault types.
(2) The collected monitoring data are typical time series, and the degradation information of the previous time steps can help predict the RUL at the current step [16]. Therefore, it is crucial to learn and extract the temporal knowledge that contains the tendency and monotonicity information of degradation processes.
(3) The normal working state of a bearing lasts for a long time, and the degradation process is usually a slow process that takes several months or even years, so it is difficult to collect a large amount of run-to-failure degradation data for model training.
(4) Industrial rolling bearings typically operate under complex working conditions, and the working conditions change frequently. Thus, effective RUL methods need to be considered under different working conditions and the inconsistent data distribution also needs to be further addressed.

Fortunately, several methods have been studied and achieved excellent performance. RUL prediction methods can be classified into three categories: physics model-based [17,18], statistical model-based [19–20] and data-driven methods [21–23]. The physics model-

based approaches focus on describing the degradation mechanism of rolling bearings to estimate and infer current and future degradation processes[24]; examples include, the Paris model [25] and the Forman model [26]. Although bearing degradation failure behaviors can be captured, these models are sensitive to working condition changes. Statistical model-based approaches are used to predict the RUL values of bearings by constructing stochastic degradation models such as the Wiener process [27–29], the Gamma process [30,31], and the inverse Gaussian process [32,33]. These stochastic models describe the uncertainty of the bearing degradation process in terms of random factors resulting from measurement errors, different loads, and momentary fluctuations. Therefore, statistical model-based solutions can effectively describe the probability distribution of RUL prediction results. However, these methods have certain limitations since specific distribution conditions need to be satisfied, and extensive expert knowledge is also needed.

With the rapid development of artificial intelligence, RUL prediction methods based on data-driven have attracted increasing attention. Data-driven methods mainly focus on constructing the mapping relationship from the monitoring data to the RUL label space. Different from traditional model-based methods, data-driven methods are highly dependent on the quality and quantity of training data rather than comprehensive prior knowledge, which offers promising value for promotion in industrial application scenarios. Various machine learning algorithms with shallow models, such as support vector machine (SVM) [34], Gaussian process regression (GPR) [35], and hidden Markov model [36] are effective RUL prediction methods. Due to their good self-adaptive feature extraction abilities [37], deep learning-based methods can effectively mine evolutionary knowledge concerning the bearing degradation process. Especially in recent years, deep neural networks like deep belief network (DBN) [37,38], long short-term memory (LSTM) [39], convolutional neural network (CNN) [40], and auto-encoder (AE) [41], have also been applied in RUL prediction and have achieved competitive results. Unfortunately, one of the major limitations is that the majority of data-driven methods generally need large amounts of run-to-failure data for model training.

As mentioned above, it is difficult to collect sufficient annotated historical data in many real-world industrial applications [23,42,43]. Furthermore, the same distribution needs to be met between training data and test data, if the degradation data acquired under different working conditions are utilized directly for model training [44–46], inaccurate prediction results will be obtained due to the violation of the independent and identical distribution criterion.

Transfer learning (TL) learns knowledge from one domain and transfers it to a different but similar domain [47,48]. This technique has received attention in text classification [49,50], computer vision [51,52], and natural language processing [53,54]. In the field of PHM, TL has been proven to be an effective tool for solving typical rolling bearing problems such as fault detection [55,56] and fault diagnosis [57–59]. Based on the fact that the intrinsic consistency exists in the degradation process of different bearings [60,202], the role of TL in RUL prediction is to extract domain-invariant temporal feature representations and transfer the degradation knowledge acquired under different working conditions via domain adaptation.

According to our literature survey, TL-based RUL prediction solutions have attracted increasing attention from both academia and industry in recent years. The commonly used TL algorithms in the RUL prediction field include model-based TL methods, discrepancy metric-based TL methods, and adversarial training-based TL methods, which will be described in detail in Section 2. Other TL techniques, such as knowledge distillation [61,62], and hybrid approaches [63,64] have also been used for RUL prediction of rolling bearings. Publications about TL-based RUL prediction for rolling bearings have been counted according to the search results from the Web of Science and ScienceDirect. The keywords that have been utilized to review publications are “transfer learning”, “remaining useful life” and “bearing”. Notably, “transfer learning” can be replaced with “different working conditions” or “cross-working conditions”. To ensure the quality of the selected papers, the search articles were thoroughly read and examined, and some duplicate or irrelevant articles were removed. Following the retrieval process, the final search results were gathered for further analysis. As shown in Fig. 1. the statistics of the results show that a new upsurge in the research on the application of TL techniques to RUL prediction is in the making, especially in the last two years. Therefore, it is necessary to survey the current literature to help scholars quickly overview the research progress. In particular, from the perspective of industrial application scenarios, engineers can select more appropriate solutions according to their specific research questions [65–67].

Some review studies have been published on the topic of RUL prediction for rolling bearings. In one of the representative works,

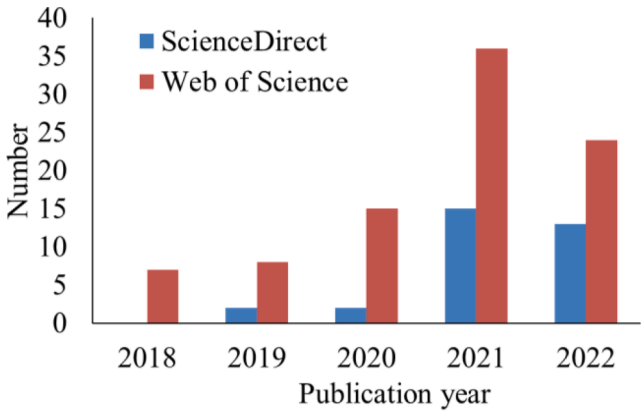


Fig. 1. Number of publications during 2018.01 ~ 2022.12 in two scientific index engines.

Cubillo et al. [68] summarized physics-based models with typical failure modes for rotating machinery. However, it mainly focused on simple formulas, and there was a lack of discussion on complex models that are more appropriate for industrial scenarios. In 2018, Lei et al. [69] provided a review covering the whole process of RUL prediction, including data acquisition, health indicator construction, health status division, and prediction model construction. The classical statistical models and artificial intelligence techniques were introduced in this review. However, it is necessary to update the state-of-the-art approaches in recent years. Kan et al. [70] comprehensively discussed different prognostics techniques applied for nonlinear and nonstationary processes up to 2015. Nevertheless, this review mainly focused on the prediction methods without a thorough analysis of different working conditions. Zhang et al. [71] presented recent developments in Wiener process-based methods for RUL estimation, as well as their applications in the field of PHM up to 2018. Although the conventional Wiener process-based models and the associated variants were systematically reviewed, this review concentrated on the same type of degradation data. A variety of data types and degradation knowledge under cross-working conditions should be considered for prognostics. Wang et al. [72] focused on RUL prediction based on various deep learning approaches, including AE, DBN, CNN, and recurrent neural networks (RNNs), and presented a brief survey in 2020. More algorithms, especially TL algorithms, should be supplemented to diversify the prediction techniques that are more applicable to practical industrial scenarios. Javed et al. [73] provided a comprehensive review of RUL prediction methods from the perspective of different prognostic techniques, evaluation criteria, and typical component-level applications up to 2017. In this review, the dynamics and principal factors in real prognostic systems were highlighted. However, it was published five years ago, and a more recent comprehensive overview needs to be provided to capture the developments over the last five years.

The above reviews discuss various RUL prognostic methods from different perspectives. However, these works have several limitations as follows:

(1) Historical review articles [68–73] focus more on physics model-based methods, statistical model-based methods, and traditional machine learning methods, and they generally ignore the TL-based methods. However, TL-based methods are extremely crucial under real industrial scenarios, especially for scenarios involving insufficient data and cross-working conditions.
(2) Most of these papers [68,70–72] mainly reviewed publications from the perspective of algorithms. However, different RUL prediction algorithms have similar or different usage scenarios, which results in algorithm-oriented review articles providing limited help and guidance when faced with practical industrial applications.
(3) The above articles [68–71,73], were published four or more years ago. In recent years, some novel RUL prediction methods, especially TL-based methods have received increasing attention. However, there is a lack of systematic reviews that include the most recent research advances.

To overcome the above limitations, this paper offers a comprehensive review of TL-based RUL prediction approaches for rolling bearings from the perspective of TL theory and industrial application scenarios. On the one hand, this paper describes the theories of TL and concentrates on the TL techniques that are frequently utilized in RUL prediction, including model-based, discrepancy metric-based, and adversarial training-based methods. On the other hand, this review reasonably divides the industrial application scenarios into three categories, i.e., target domain has run-to-failure data, target domain has no run-to-failure data, and target domain has fault information. The corresponding solutions in each category are summarized and discussed in detail.

Compared to the existing review papers, the significant contributions of this review are as follows.

(1) TL theory and typical TL techniques related to RUL prediction are summarized into three categories, providing a basic tutorial and framework for beginners to quickly grasp the related work developments from the algorithmic perspective.
(2) The TL-based RUL prediction solutions are comprehensively evaluated and systematically analyzed according to three typical industrial scenarios, yielding suggestions for engineers to select appropriate algorithms based on their research requirements.
(3) Challenges and future trends in the RUL prediction fields are discussed, aiming to put forward valuable and promising research directions for scholars.

This paper is organized as follows. Section 2 describes the basic definitions of TL-based RUL prediction and provides typical TL techniques in RUL prediction. Section 3 discusses the published literature in detail from the perspective of practical industrial scenarios. The current challenges and future trends in RUL prediction are presented in Section 4. Finally, Section 5 concludes the review.

2. TL theory in RUL prediction

Before reviewing the solutions in detail, the problem definition and general procedure of TL-based RUL prediction are first introduced. Then, the TL techniques that are commonly used in RUL prediction are summarized from the perspective of algorithms, which helps readers quickly understand TL theory and the developments related to the RUL prediction problem.

2.1. Problem definition of TL-based RUL prediction

In industrial scenarios, it is difficult to collect a large amount of labeled data for training an RUL model under specific working conditions. Therefore, traditional data-driven methods have failed to construct an accurate RUL prediction model in this case. With the help of TL, the problem can be solved by borrowing some sample data obtained under other working conditions. In recent years, research on TL-based RUL prediction has received increasing attention in the industry. This subsection presents the problem definition

of TL-based RUL prediction, including the concepts adopted in TL, the prediction task and the role of TL.

Denote by $\{x_i^S\}_{i=1}^{n_s}$ the run-to-failure data obtained under a certain working condition, where n_s is the number of samples. x_i^S belongs to the sample space χ^S, which has a marginal probability distribution $P(\chi^S)$. Denote by $\{x_i^T\}_{i=1}^{n_t}$ the monitoring data obtained under another working condition, where n_t is the number of samples. x_i^T belongs to the sample space χ^T with a marginal probability distribution $P(\chi^T)$. Data acquired under different working conditions have diverse distributions because of many aspects, including differences in assembly and measuring operations, i.e., $P(\chi^S) \neq P(\chi^T)$.

Referencing the concept of TL [74], the preconditions for performing RUL prediction under different working conditions can be described as follows.

(1) All bearings have a similar degradation process from a healthy state to complete failure [75]. Although the degradation trends of various bearings appear to vary, the intrinsic similarity of degradation processes under cross-working conditions is assumed [75–77].
(2) The source domain D^S consists of χ^S and $P(\chi^S)$, i.e., $D^S = \{\chi^S, P(\chi^S)\}$. This domain provides prior knowledge about degradation information for RUL prediction.
(3) The target domain D^T consists of χ^T and $P(\chi^T)$, i.e., $D^T = \{\chi^T, P(\chi^T)\}$. It should be noted that the target domain usually has only a small amount of run-to-failure data or even no whole-life data, which is insufficient for the construction of an RUL prediction model.

Mathematically, RUL prediction is essentially a time series regression problem that finds a mapping relationship from χ^S to the RUL label space γ^S, i.e., $f: \chi^S \mapsto \gamma^S$. As mentioned above, it is difficult to directly construct an RUL prediction model in the target domain: $f: \chi^T \mapsto \gamma^T$. The strategy of TL involves improving the RUL prediction performance in the target domain by transferring the degradation information learned from the source domain data.

2.2. TL-based RUL prediction procedure for rolling bearings

As shown in Fig. 2, the general procedure of intelligent TL-based RUL prediction can be divided into four steps: (1) collect data under different working conditions from machinery sensors and divide the dataset into training and testing datasets; (2) extract features using the training dataset; (3) construct an intelligent neural network model based on the training dataset, and (4) calculate the prediction results using testing data by employing the developed regression model.

***(1) Data acquisition*:** The first step in RUL prediction is data acquisition, which provides basic monitoring information for subsequent processes. Generally, rolling bearings exhibit in healthy states for a long time, so it is time-consuming and difficult to gather a large amount of run-to-failure data for model training. In TL, run-to-failure data collected under one working condition (source domain) can help with modeling test data, which are collected under another working condition (target domain) and generally do not

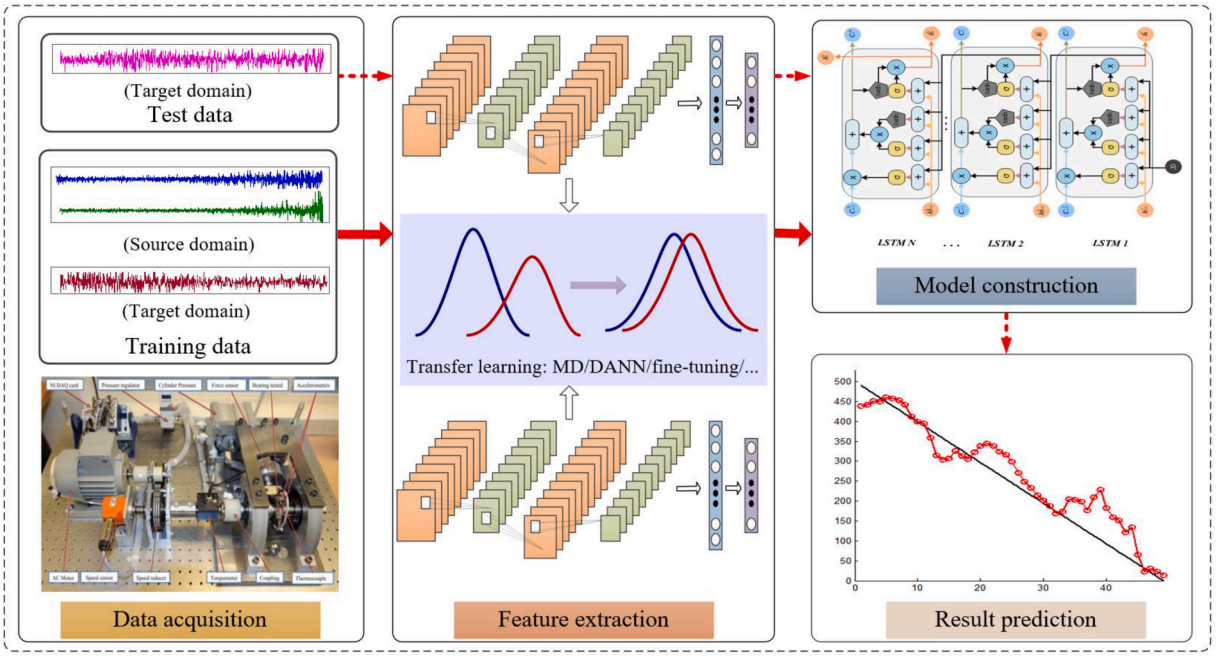


Fig. 2. Intelligent TL-based RUL prediction process.

provide enough run-to-failure data.

***(2) Feature extraction*:** Feature extraction is expected to derive useful fault knowledge that can reflect the degradation process of rolling bearings from the raw data, and the quality of the obtained features determines the accuracy of RUL prediction to some extent. Common statistical features include time-domain features, frequency-domain features, and time–frequency-domain features. In recent years, deep learning methods have also been widely used in feature extraction tasks and can obtain better results than traditional manual feature extraction techniques.

***(3) Model construction*:** Model construction is an extremely important step for RUL prediction. Various prediction models have different application ranges and produce varying prediction results. Generally, typical deep neural networks include feedforward neural networks (FNNs) [78–80] (such as DBN and CNN) and RNNs [81–83]. FNNs can directly construct a mapping relationship from the sample data to RUL values, and the training process is effective and time-saving. One of the major limitations, however, is that detailed temporal information cannot be adequately considered. In contrast, RNNs reconstruct samples with time windows; thus, temporal knowledge can be implicitly expressed during model training. Although severe gradient vanishing and exploding problems [84] make the optimization process of RNNs more challenging than that of FNNs, these approaches are becoming increasingly popular in the field of RUL prediction with the advantage of not requiring expert knowledge.

***(4) Result prediction*:** Based on the above three steps, a well-trained RUL prediction model can be constructed using the given training data, and the predictive result of testing data can be obtained by feeding the testing data into the well-trained RUL prediction model.

Usually, the data distributions of the training data in the source domain and the testing data in the target domain are inconsistent. The role of TL is to extract domain-invariant features via domain adaptation in the second step or to construct a parameter-shared RUL prediction model in the third step.

2.3. Typical TL techniques in RUL prediction

With the development of artificial intelligence techniques, a variety of TL methods are constantly being studied. There is no unified standard or criterion for the categories of TL. In this section. We summarize three categories of TL techniques that are commonly used in RUL prediction from the perspective of algorithms, including model-based TL methods, discrepancy metric-based TL methods, and adversarial training-based TL methods, which can provide a basic research framework for beginners in academic studies to better understand the current research works.

2.3.1. Model-based TL methods

Model-based TL methods assume that individual models for both source and target domains should share some parameters or prior distributions of the hyperparameters [74]. As an important branch of model-based TL methods, the fine-tuning strategy transfers the network parameters from the source domain to the target domain when two domains are related [60,85,86]. By learning shared parameters and prior knowledge, a well-trained network structure on the source domain can be quickly applied to test data from the

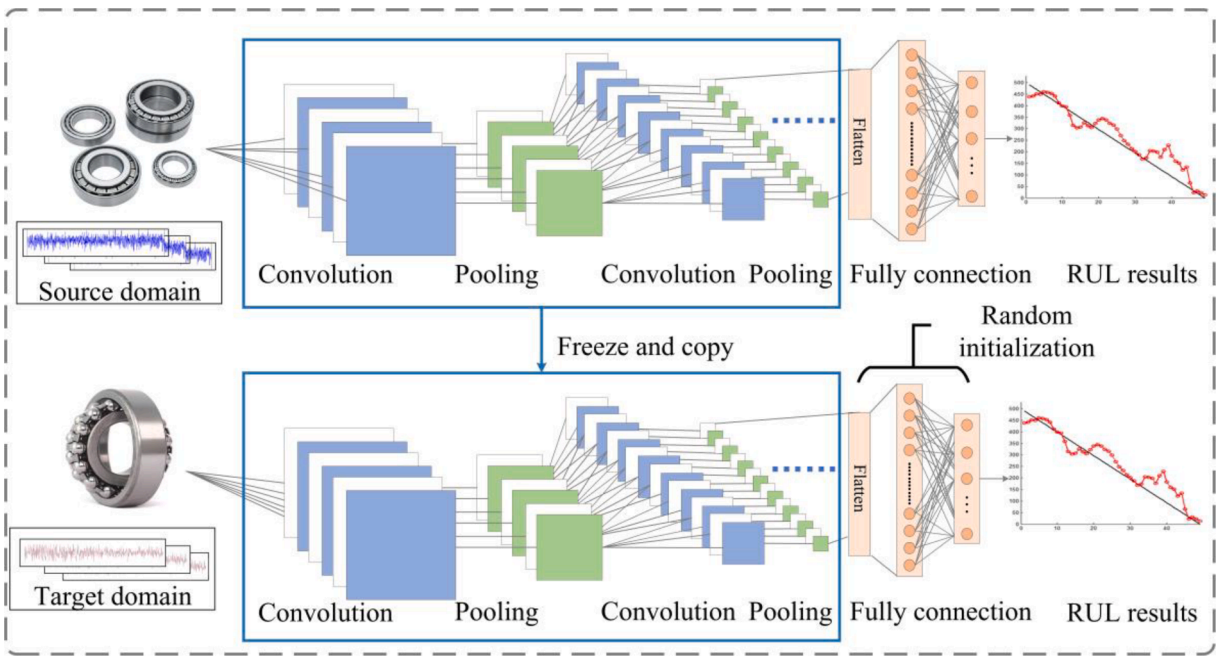


Fig. 3. Fine-tuning structure.

target domain. The fine-tuning process is presented in Fig. 3. It consists of two parts: (1) pre-training a deep neural network (such as CNN) on the source domain and saving the best model parameters and (2) freezing and copying the model parameters of the hidden layers in the source domain as the initialization parameters of the model in the target domain, and randomly initializing the parameters of a fully connected layer. Then, the model parameters are further fine-tuned with target domain data. In RUL prediction, the source and target domains represent the collected degradation data under different working conditions.

In the field of RUL prediction, source domain models are often trained with large amounts of run-to-failure data under one working condition, and the target domain usually contains only a small amount of data. There are two strategies for fine-tuning a model: the first is freezing all the weight parameters of the pre-trained network and updating only the weights of the fully connected layer, which is suitable for the case where the target domain has a small amount of data, but is sufficiently similar to the source domain. The second is fine-tuning the parameters of certain layers by fixing the weights of the lower layers of a network and only updating the parameters of the higher layers, which can avoid the overfitting problem caused by directly training a model with a small amount of data from the target domain. Depending on the quantity and quality of the target domain data, the selective feature layer learning strategy is more effective for different tasks [87–89]. The lower-level feature refers to features obtained in the first few layers of the deep neural network, which represent more general features and have good generalization for other tasks, while the high-level features are semantic features that are more relevant to specific tasks [90–92]. For the details of the training process, please refer to [93].

2.3.2. Discrepancy metric-based TL methods

Discrepancy metric-based TL methods are committed to minimizing the feature distribution discrepancy between the source and target domains, as shown in Fig. 4. For the RUL prediction of bearings, a large feature distribution discrepancy exists among different domains because the data are collected under different working conditions. Fortunately, the discrepancy can be reduced as much as possible by minimizing the distribution distance, which is calculated using some discrepancy metrics. Based on the common feature spaces, the RUL prediction model trained on the source domain is also applied to the target domain data.

The distribution discrepancy can be calculated by typical discrepancy metrics such as maximum mean discrepancy (MMD) [94], Kullback-Leibler (KL) divergence [95], Jensen-Shannon divergence [96], and Euclidean distance [97]. The MMD distance is a popular nonparametric criterion for calculating the discrepancy between two feature distributions [98]. It can avoid calculating the intermediate density and reduce the calculation workload [75]. The MMD distance is defined as [99]:

$$L_{MMD} = MMD(X^s, X^t) = \left\| \frac{1}{n^s} \sum_{i=1}^{n^s} \varphi(x_i^s) - \frac{1}{n^t} \sum_{j=1}^{n^t} \varphi(x_j^t) \right\|_H \tag{1}$$

where $\mathscr{H}$ denotes the Reproducing kernel Hilbert space, n^s and n^t indicate the number of features in the source and target domains, respectively. Benefitting from the advantage of not requiring label information, discrepancy metric-based TL methods can only achieve domain adaptation at the feature level. Therefore, this strategy is becoming increasingly common in RUL prediction.

2.3.3. Adversarial training-based TL methods

In addition to discrepancy metric-based methods, the adversarial training mechanism is another often used algorithm in unsupervised TL. Gannin et al. [100] first presented the domain-adversarial training of neural networks (DANN) in 2016. Inspired by generative adversarial networks (GANs), the adversarial training mechanism aims to match the feature space distribution by modifying the feature representation itself rather than by performing geometric transformation [100]. The structure is shown in Fig. 5.

The network in Fig. 5 mainly includes three parts. The feature extractor $G_f(\cdot;\theta_f)$, shown in the blue part, is designed to extract deep features from monitoring data under the source and target domains. The red part is the regressor $G_y(\cdot;\theta_y)$, which is used to receive the extracted features for the RUL prediction task. The domain classifier $G_d(\cdot;\theta_d)$, depicted in the purple portion, serves to identify which

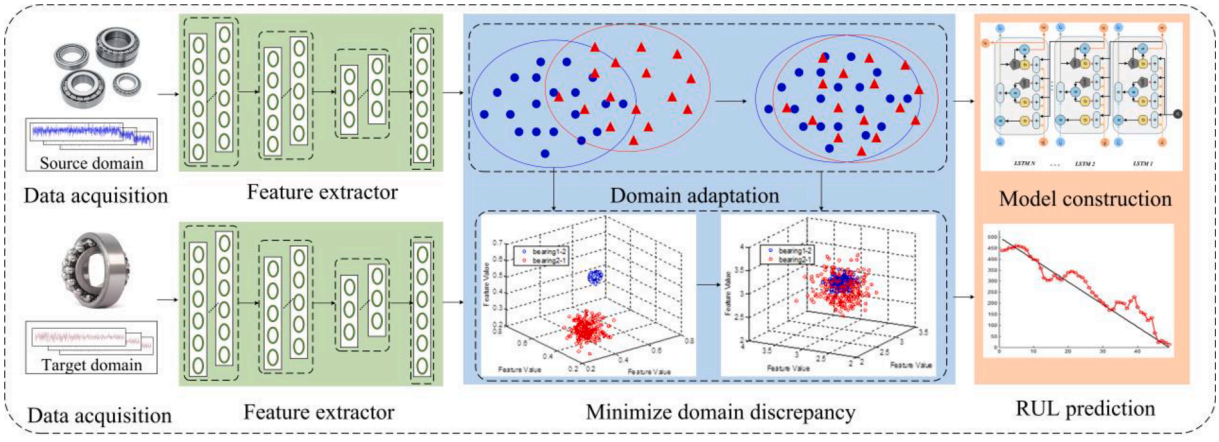


Fig. 4. Discrepancy metric-based TL methods.

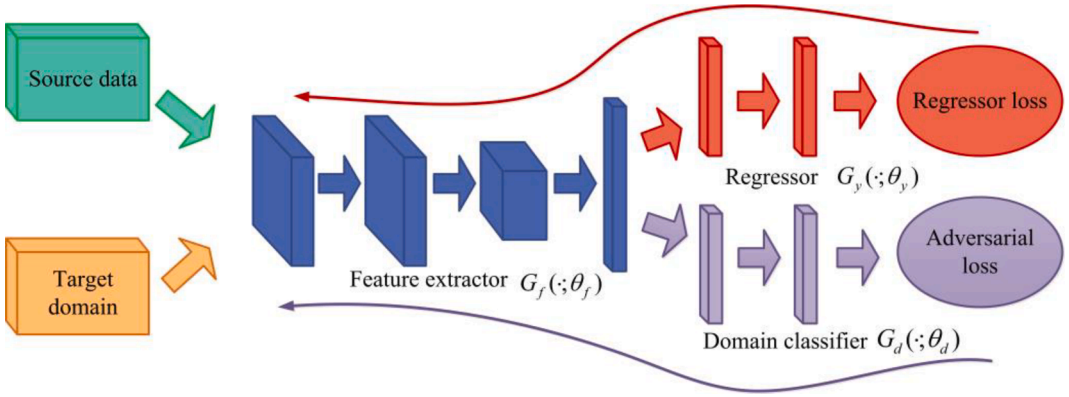


Fig. 5. Network structure of adversarial training.

domain the training data come from.

The DANN optimizes the parameters θ_f and θ_y of G_f and G_y by minimizing the task loss and feature loss, respectively:

$$(\widehat{\theta}_f, \widehat{\theta}_y) = \text{argmin} E(\theta_f, \theta_y, \theta_d) \tag{2}$$

Then, the DANN optimizes the parameters θ_d by maximizing the loss of the domain classifier G_d:

$$(\widehat{\theta}_d) = \underset{\theta_d}{\text{argmax}} E(\theta_f, \theta_y, \theta_d) \tag{3}$$

Similar to the GAN training procedure, the two processes are alternated until the network converges.

For implementation convenience, a gradient reversal layer is incorporated into the backpropagation module of the network. Therefore, the above two processes can be combined into a single objective function:

$$E(\theta_f, \theta_{y'}, \theta_d) = \sum_{x_i \in D_s} L_y(G_y(G_f(x_i)), y_i) - \lambda \sum_{x_i \in D_s \cup D_t} L_d(G_d(G_f(x_i)), d_i) \tag{4}$$

where L_y is the loss of $G_y(\cdot;\theta_y)$, L_d is the loss of $G_d(\cdot;\theta_d)$, and d_i is the domain label. When the data comes from the source domain, $d_i = 0$; otherwise, $d_i = 1$.

In addition, dynamic adversarial adaptation networks [101], adversarial discriminative domain adaptation [102], Wasserstein GAN [103], and CatGAN [104] have significant impacts on reducing distribution discrepancies.

3. RUL prediction based on TL for different application scenarios

In practical industrial scenarios, accurately assessing the remaining life can help engineers and decision-makers promptly manage the underlying risk caused by unexpected failure. Following the above motivation, some excellent literature reviews [65,105–107] have been developed on RUL prediction from the perspective of industrial applications to further develop real engineering-oriented prognostics methods. For example, Sikorska et al. [65] presented a comprehensive review of industrial application-oriented prognostics, including existing failure mode prognostics, future failure mode prognostics, and post-action prognostics. The corresponding theoretical limitations and influencing factors of different prognostic methods were also analyzed in detail. Scarf et al. [105] further highlighted that the mathematical modeling of maintenance should be driven by applications where real maintenance-related problems and typical maintenance models have been thoroughly discussed. A systematic review was presented by Engel et al. [106] for practical issues in RUL estimation, and the underlying relationships among the precision, accuracy, and uncertainty for RUL prediction were emphasized. Although useful in understanding the degradation state and failure behaviors, they still require sufficient engineering experience and do not consider the quantity and quality of monitoring data in industrial applications.

From the perspective of the amount and diversity of practical data in industrial applications, TL-based RUL prediction methods for different application scenarios are compressively summarized in this section. Specifically, according to the availability of run-to-failure data, industrial scenarios can be divided into RUL prediction with run-to-failure data and RUL prediction without run-to-failure data. Compared to only exploiting degradation information from data perspective, some typical fault information, such as fault types and health status, can assist in improving the accuracy and reliability of RUL prediction in industrial applications; therefore, RUL prediction with fault information is introduced as a third scenario. These categories cover almost all industrial scenarios. Furthermore, the historically published literature in each scenario is thoroughly analyzed and evaluated. This new taxonomy is expected to provide helpful references for operators and engineers when choosing appropriate solutions according to their specific working scenarios.

3.1. Different scenarios characteristics and goals

Depending on whether the target domain has run-to-failure auxiliary data engaged in model training, RUL prediction tasks in industrial scenarios can be classified into RUL prediction with run-to-failure data and RUL prediction without run-to-failure data. The characteristics of different industrial scenarios determine the applicable scope of different TL methods. In addition, some solutions yield improved RUL prediction performance with the help of fault information. An illustration of the three application scenarios is shown in Fig. 5, and brief introduction are as follows.

1) ***RUL prediction with run-to-failure data*:** As shown in Fig. 6(a), the run-to-failure auxiliary data in the target domain are available, which means that the data in the target domain can be fully utilized to construct an RUL prediction model through TL methods. The

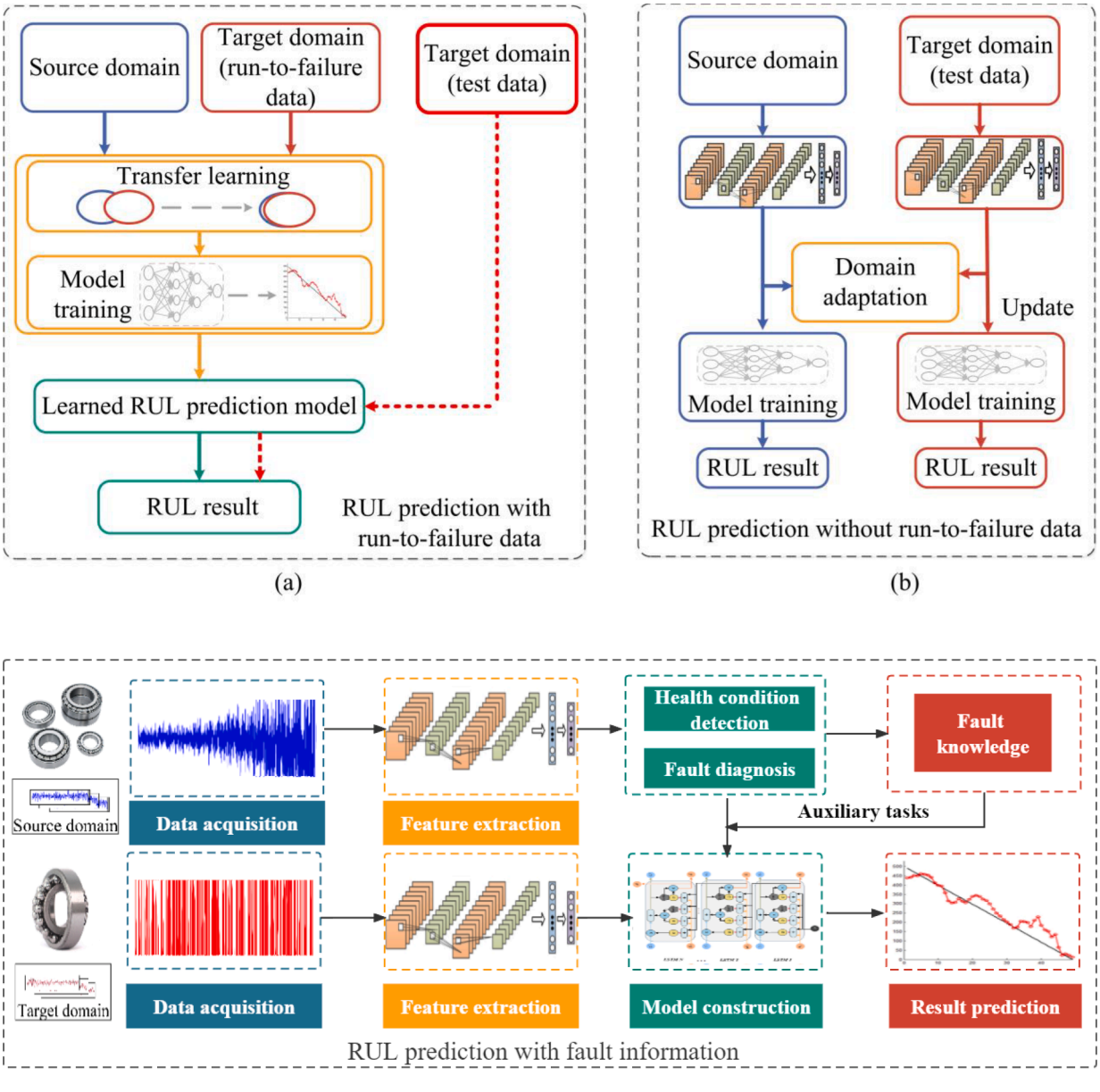


Fig. 6. Flowcharts of three industrial application scenarios for RUL prediction (a) RUL prediction with run-to-failure data, (b) RUL prediction without run-to-failure data, and (c) RUL prediction with fault information.

RUL values of the test bearing can be directly obtained by feeding data into the well-trained RUL prediction model. In such a scenario, the goal is to improve the robustness and generalization of the prediction model.

2) ***RUL prediction without run-to-failure data*:** Another popular scenario in industrial applications is one in which the target domain lacks additional run-to-failure data, and the data arrives sequentially. In such a scenario, the key to performing the RUL prediction task is borrowing the temporal degradation information gained from the source domain and using unsupervised TL strategies to adapt domain-invariant feature spaces. Borrowing the domain-invariant features, an accurate RUL prediction model can be subsequently constructed. Furthermore, the model is dynamically updated with online data. The flowchart is illustrated in Fig. 6(b), and the main goal is to learn degradation information and accurately predict RUL values.
3) ***RUL prediction with fault information*:** In addition to the two typical scenarios mentioned above, we also note that different degradation processes exist in different fault types or health condition stages, which can provide prior knowledge for accurate RUL prediction, as shown in Fig. 6(c). The ultimate goal in this scenario is to enhance the reliability and practicality of RUL prediction models with the help of fault information.

Currently, data in case studies have mainly been collected in the laboratory environment. The commonly used rolling bearing datasets include the accelerated life testing bearing dataset [108], IEEE PHM 2012 Challenge bearing dataset [109], XJTU-SY bearing dataset [110], CWRU dataset [111], IMS bearing dataset [112], and bearing datasets collected under time-varying rotational speed conditions [113]. However, limited by industrial scenarios where failure bearings are not allowed to work for a long time, few case studies demonstrate the application that has been published in the literature in the practical operational environment. Therefore, more work needs to be completed to verify the accuracy and reliability of prognostic methods in real industrial applications.

3.2. RUL prediction with run-to-failure data

For the scenario of RUL prediction under cross-working conditions, many TL-based RUL prediction solutions have been investigated to improve the generalization performance of prediction models in the target domain. It should be noted that some statistical model-based solutions can also solve this problem by using parametric working conditions [114–118], which break through the limitations of traditional statistical model-based methods that construct specific prediction models for different working conditions. The advantage of statistical model-based solutions is that they do not require a large amount of data and yield interpretable prediction results [119,120]. However, an obvious bottleneck behind these methods is that they generally require extensive expert knowledge to design stochastic degradation models and manually extract tendency features that are consistent with the degradation trajectory. Therefore, how to extract effective features from undesired interference noise is very important in these solutions. To overcome the limitations of statistical models, more effective data-driven TL solutions have been studied in recent years.

In this review, the existing TL-based RUL prediction solutions with run-to-failure data are divided into model-based TL solutions, discrepancy metric-based TL solutions, domain adversarial-based TL solutions, and other TL-based solutions. Table 1 provides an overview of TL-based RUL prediction solutions with run-to-failure data.

3.2.1. Model-based TL solutions

Model-based TL strategy, one of the major TL solutions for RUL prediction, pre-trains a deep network on the source domain, and then the learned model parameters are further adjusted using a small number of labeled auxiliary samples in the target domain. By sharing the neural network structure or parameters between source and target models, model-based TL solutions can improve the generalization performance of prognostics models when the run-to-failure data in the target domain are available. According to the results of the literature review, model-based TL techniques have achieved considerable success in RUL prediction under cross-working conditions.

For instance, Zhang et al. [60] proposed a TL framework for the RUL prediction of bearings. The framework includes two bidirectional LSTM (Bi-LSTM) networks. The top Bi-LSTM model was trained using data from the source domain, and the model parameters were considered the initialization of the bottom Bi-LSTM model, which was trained with run-to-failure auxiliary data in the target domain data. This work first proposed that TL can improve the accuracy of RUL prediction when insufficient labeled data are available in the target domain. A deep TL network based on a sparse autoencoder (SAE) was developed by Sun et al. [121] for RUL prediction under cross-working conditions. In the proposed method, three transfer strategies, weight transfer, TL for hidden features, and weight updating, were used to transfer a well-trained SAE network from the source domain to the target domain. At the same time,

Table 1
TL Solutions for RUL Prediction with Run-to-failure Data.

Application scenarios	Solutions	References
RUL prediction with run-to-failure data	Model-based	Zhang *et al.*[60], Sun *et al.* [121], Huang *et al.* [122], Zhang *et al.* [85]
	Discrepancy metric-based	Cheng *et al.* [123], Mao *et al.* [124], Xu *et al.* [125], Cao *et al.* [126], Rathore *et al.* [127], Wu *et al.* [128], Zhuang *et al.* [129], Ding *et al.* [130]
	Domain adversarial-based	Miao *et al.* [134], Hu *et al.* [135], Wang *et al.* [136], Zhuang *et al.* [137], Li *et al.* [138], Zou *et al.* [139], Lu *et al.*[140], Fu *et al.* [141], Miao *et al.* [142], Cheng *et al.* [143]
	Other TL methods	Suh *et al.* [144], Cheng *et al.* [145], Xu *et al.* [146], Fan *et al.* [147], Chen *et al.* [148], Hu *et al.* [149], Mao *et al.* [88], Huang *et al.* [150], He et al. [151], Zhuang *et al.* [152], Zou *et al.* [153], Cui *et al.* [154], Liao *et al.* [155], Li *et al.* [156].

the domain discrepancy was further diminished by adding KL divergence regularization. The evaluation results indicated that the proposed TL strategy is an effective tool for RUL prediction. By borrowing the source domain samples, Huang et al. [122] fine-tuned a well-trained depth-wise separable convolution recurrent network. Compared to other methods, minimal error was observed using the proposed fine-tuning technique. Zhang et al. [85] constructed a transferable RUL prediction model based on a pre-trained LSTM network, which exploits the degradation knowledge learned from the source domain to improve the generalization performance of RUL prediction in the target domain. The effectiveness of the proposed method was verified by bearing and gearbox datasets.

The success of the fine-tuning strategy has enabled model-based TL solutions to quickly predict the remaining life of bearings under cross-working conditions by transferring the obtained model parameters from the source domain to the target domain. However, a key issue that cannot be ignored is that the fine-tuning strategy requires labeled training data in the target domain for model training, which seriously limits the application scopes of these solutions because run-to-failure data cannot be collected in advance. Therefore, there is an urgent need to develop unsupervised TL solutions to tackle this problem.

3.2.2. Discrepancy metric-based TL solutions

In the field of RUL prediction with run-to-failure data in the target domain, domain adaptation is another popular TL strategy. By minimizing the distribution discrepancy between the source domain and the target domain, the feature distribution spaces in the two domains are as consistent as possible. From the perspective of discrepancy metrics, a variety of metrics such as the MMD and its variants have been adopted to calculate distribution distances [123–130].

As a pioneering work, Mao et al. [124] developed an RUL prediction model based on a least-square SVM (LS-SVM) and transfer component analysis (TCA). The TCA algorithm can effectively find a common subspace where the given feature representations are close to each other. Then, a feature space-based LS-SVM was developed for RUL prediction under different working conditions. In this way, the prediction model can be used in both the source domain and the target domain. The promising prognostic results validated the effectiveness of the proposed TL strategy. Similarly, Xu et al. [125] adopted the MMD distance to calculate the feature space discrepancy between the source domain and the target domain. Then, a Bi-LSTM network was constructed for RUL prediction. In addition, a comprehensive feature evaluation index consisting of dynamic time warping and the Wasserstein distance measure was proposed by Cao et al. [126], where the index can be used to choose domain-invariant features with good temporal characteristics. Furthermore, the MMD algorithm was adopted to address the distribution discrepancy. The experimental results showed that discrepancy metric-based solutions are valid for RUL prediction under different working conditions.

Two main problems are encountered when using the standard MMD algorithm: (1) Only one concrete kernel function, such as a Gaussian kernel or linear kernel, is used when calculating distribution discrepancies. In most cases, the kernel function is chosen manually. Therefore, selecting a reasonable kernel function is a challenging problem. (2) When there are multiple sub-source domains, it is not sufficient to align the distribution at the global scale by minimizing the MMD distance between the source domain and the target domain while ignoring the fine-grained degradation information in the sub-source domains. To address these issues, many solutions based on optimized MMD measures have been studied to enrich algorithm diversity and further improve the accuracy of RUL prediction under complex working conditions.

Some MMD variants, which are more robust than standard MMD algorithms, have been adopted to learn common domain-invariant feature representations [123,127–129]; examples include multi-kernel MMD (MK-MMD) and local MMD. To accurately predict the RUL value of rolling bearings with multiple failure behaviors, Cheng et al. [123] developed a transferable CNN model, in which the CNN was employed to learn the degradation feature of different failure behaviors, and the MK-MMD was combined into the optimization function to minimize the feature space distribution discrepancies among various fault behaviors. The proposed method achieved better RUL prediction results than the standard MMD algorithm. However, it was performed under the same working conditions. Aiming to improve the prediction performance achieved under different working conditions, Zhuang et al. [129] presented a transferable cross-domain RUL estimation approach. A novel domain adaptation technology with a contrastive loss and the MK-MMD was designed to learn domain-invariant features. Additionally, a temporal convolution network and a residual self-attention mechanism were introduced to extract degradation features. The promising obtained prognostic results validated the effectiveness of the proposed TL strategy. Different from the MK-MMD, the local MMD considers the relationship between the subdomains in the source domain. Ding et al. [130] proposed a novel TL method, called the deep subdomain adaptive regression network, for predicting the RUL values of rolling bearings across working conditions. In the proposed method, based on the synergy of an auxiliary category label, an auxiliary subdomain classifier, and the local MMD, the distribution discrepancies between the subdomains in the source and target domains can be aligned simultaneously. Finally, the kernel regression method was used to predict RUL values. This work was the first attempt to achieve fine-grained subdomain adaptation for regression models.

The main advantage of the above solutions is that the distribution discrepancies between the feature representations of the source and the target domains can be reduced without the label space information. The above solutions realize domain adaptation at the feature level and obtain effective predictive results under different working conditions. However, they may cause negative transfer in complex and harsh environments because they have a potential limitation that the target domain and the source domain are very similar. Therefore, more robust TL methods need to be developed to further improve the reliability of RUL prediction in industrial scenarios.

3.2.3. Domain adversarial-based TL solutions

In addition to discrepancy metric-based solutions, adversarial training mechanisms [131] have shown excellent domain adaptation performance [100,132]. The core of domain adversarial-based solutions is to learn a domain-invariant feature space in which the domain classifier cannot distinguish which domain the input data come from [133]. Based on common feature spaces, an RUL

prediction model can be constructed and achieve good prediction results in both the source domain and target domain. In recent years, domain adversarial training has attracted considerable attention, and many related research papers have been published in the field of RUL prediction under different working conditions [134–139].

For example, Wang et al. [136] constructed a dynamic adversarial adaptation network (DAAN) for rolling bearing RUL prediction under cross-working conditions. Compared with the single domain classifier in the DANN, global and local domain classifiers were introduced simultaneously to the DAAN to reduce the distribution differences between the source and target domains. The experimental results showed that the proposed method was more competitive than the traditional RUL prediction method. Zhuang et al. [137] proposed a metric-based adversarial domain adaptation approach with a supervised positive contrastive module and a dual self-attention module, which not only effectively learned domain-invariant degradation features but also achieved good prediction performance when the labeled training data were insufficient in the target domain. Extensive experiments were conducted in twelve cross-domain scenarios for two bearing cases. In addition, Li et al. [138] further provided a TL-based RUL estimation method for different machines. To improve the model generalization performance, adversarial training was also introduced to align the domain-invariant feature spaces of both healthy state and degradation state data. Zou et al. [139] developed a multi-domain adversarial network based on a feature coding layer to improve the performance of feature transfer. The results demonstrated that the proposed method could offer a novel and promising tool for RUL predictions under varying working conditions.

According to the aforementioned analysis, the majority of the proposed solutions are based on either discrepancy metrics or adversarial training mechanisms. Discrepancy metric-based methods and domain adversarial-based methods implement domain adaptation from two different perspectives. Research has shown that better prediction results can be obtained by using both strategies simultaneously. Therefore, it is significant to realize domain adaptation by combining distribution discrepancy and adversarial training mechanisms, which can make full use of their advantages to further improve the RUL prediction performance under different working conditions [140].

For example, a deep residual LSTM network with domain-invariance was proposed by Fu et al. [141]. In the proposed method, two strategies were simultaneously utilized to learn domain-invariant features between the source and target domains. The first approach was domain adaptation, which reduced the domain distribution discrepancy by adding MK-MMD constraints. The second technique was domain confusion, which further learned shared feature spaces through adversarial training. The excellent performance of the proposed method was verified on two datasets. Combining an adaptive feature extractor-based adversarial training mechanism and the MMD distance measure, Miao et al. [142] constructed a sparse domain adaptation network (SDAN) for bearing RUL prediction under different working conditions, where the feature distribution shift across domains was reduced as much as possible. An analysis of the experimental results confirmed that the proposed SDAN is an effective tool for bearing RUL prediction under cross-working conditions. Similar to the work of Ding et al. [130], Cheng et al. [143] developed a dynamic domain adaptation approach for RUL prediction under multiple sub-source domains, where reverse validation technology was utilized to select appropriate source samples for assisting the RUL prediction model training process. In addition, dynamic domain adaptation was employed to reduce the feature distribution inconsistency caused by different working conditions.

Domain adversarial-based TL solutions aim to extract domain-invariant features between the source and target domains by the zero-sum game of feature extractor and domain classifier. Benefiting from the strength of adversarial strategy, effective RUL prediction of bearings can be achieved in the industrial scenario, where only monitoring data can be collected in the target domain. Although domain-adversarial-based TL solutions have achieved significant success, a major drawback of domain adversarial-based solutions is that the adversarial training process requires more computational resources than other transfer learning algorithms, which limits their applicability in complex industrial scenarios to some extent. Further work should be conducted to meet the actual needs of the industry.

3.2.4. Other TL-based solutions

Numerous studies have been conducted on RUL prediction under different working conditions using discrepancy metric-based and domain adversarial-based TL algorithms in recent years. Moreover, some other reported methods have improved the effectiveness and generalization performance of TL-based RUL prediction models [144–156].

For instance, Fan et al. [147] constructed a consensus self-organizing model to extract transferable features for building an RUL regression model. Chen et al. [148] introduced a principal curve in the source domain to first represent the degradation process, and then target domain features similar to the principal curve were selected to construct a prediction model. The experimental results showed that the proposed method is suitable for RUL prediction under different working conditions. Considering the inherent properties of different domains, Hu et al. [149] proposed a deep feature disentanglement-based TL network for rolling bearing RUL prediction under cross-working conditions. The advantage of this network is that it can simultaneously learn the shared domain-invariant features and private features through a pair of joint learning autoencoders. To improve the generalization ability of the prediction model, a transferable prediction model based on the working condition's common benchmark was constructed by Li et al. [156]. Furthermore, Mao et al. [88] developed a novel deep transfer learning approach from the perspective of sample interpretability. In the proposed method, a transfer domain validity index was created to quantify the contribution of every sub-source domain according to the observed degradation tendency similarity. Finally, a new stacked LSTM network with selective TL was constructed for RUL prediction under cross-working conditions. The experimental results proved that the proposed approach could help transfer useful degradation knowledge from the source domain to the target domain, and further improve the RUL prediction performance.

Based on the literature surveyed above, TL methods have made significant breakthroughs in RUL prediction for rolling bearings under different working conditions. However, a key hypothesis in these studies is that several run-to-failure training data are available in the target domain. These studies are not suitable for scenarios where run-to-failure data are not available in the target domain.

3.3. RUL prediction without run-to-failure data

Compared to the application scenario in which the target domain has run-to-failure data, some rolling bearings work under new operating conditions, and it is not easy to collect run-to-failure data for training a transfer learning model. There are two main difficulties in solving RUL prediction in such scenarios. The first difficulty is how to learn degradation knowledge from a small amount of unlabeled data. The second difficulty is that the constructed prediction model needs to be updated constantly using sequentially-arriving data. Fortunately, some works have attempted to address these problems, and Table 2 summarizes the current TL-based solutions.

3.3.1. Domain adversarial-based TL solutions

In practical industrial applications, it is quite challenging to bridge the domain gap between extensive labeled data in the source domain and a small amount of unlabeled training data in the target domain. In such cases, domain adversarial-based solutions have been proven to present excellent performance for RUL prediction without run-to-failure data [43,157–160].

As a classic work, an unsupervised deep domain adaptation strategy was initially introduced by Costa et al. [43]. In this study, a DANN model was applied to learn shared feature representations between annotated data in the source domain and unlabeled data in the target domain, which is very important for domain adaptation. A shared LSTM network was executed to extract temporal-dependent features and improve the accuracy of online RUL prediction. To address the issue that the target domain only has samples during the initial degradation stage, Shahin et al. [158] developed a deep TL approach based on DANN, where a consistency-based regularization term was adopted to eliminate the negative effect caused by incomplete data from the target domain and ensure the accuracy and reliability of the prediction model. Mao et al. [159] presented a self-supervised learning strategy for online RUL prediction. By learning degradation knowledge from offline data, and using pseudo-label skills, this approach transformed the unsupervised learning problem into a self-supervised learning problem, which has achieved reliable RUL prediction without run-to-failure data in the target domain. Distinguishing from the above methods of considering only single-source domain adaptation, Ding et al. [160] developed a multisource domain adaptation network for RUL prediction under different working conditions, in which a two-stage strategy of domain-specific distribution adaptation and domain-specific regressor adaptation was employed to make high-performance predictions.

These solutions attempt to solve the RUL prediction in the target domain using the regressor trained with labeled data in the source domain after adapting the feature spaces, and have achieved excellent results. Nevertheless, these studies are still in the initial stage and more advanced TL strategies are expected to be investigated in the future.

3.3.2. Other TL-based solutions

In addition to the abovementioned TL methods, some other methods have been explored for solving the problem of a lack of run-to-failure data in the target domain. To perform online prognostic tasks, Zeng et al. [161] constructed a two-phase model with an offline stage and an online stage. To align the feature spaces of the source and target domains, a CNN model was built during the offline stage utilizing a semisupervised learning method. In the online stage, only early fault data in the target domain were used to update the model parameters. Thus, the prediction model requires only a few iterations for online RUL prediction. The experiment showed that, compared with other TL methods, the proposed method further aligned the feature spaces of the different domains and improved the RUL prediction results. Aiming at minimizing the feature distribution difference between the source domain and the target domain, Wang et al. [162] employed an RUL prediction framework, where a derived training set was generated by feeding data from the target domain into the prognostics model in the source domain. Then, the training set was input into the improved temporal convolution network to generate a transfer prediction model. Ding et al. [163] employed a new TL network for domain adaptation. Combining the MMD distance, correlation alignment, and a domain classifier, the network can minimize the first-order and second-order domain distribution discrepancies between the source and target domains, and the network performance has been validated on a sapphire-bearing bush fatigue test-rig dataset. Additionally, Wang et al. [164] developed a novel joint dictionary matrix factorization-based TL strategy for obtaining a shared latent feature space that is independent of the working conditions. Then, a regressive CNN model was constructed based on the shared feature space for performing bearing RUL prediction under different working conditions. Zhang et al. [44] proposed a subspace representation regularization strategy for realizing RUL prediction. Through healthy state alignment and degradation direction alignment, this method can reduce the domain gap even if the target domain has no run-to-failure monitoring data. Extensive experiments conducted on the XJTU-SY dataset validated the reliability of the proposed strategy, which presents a promising solution for the prognostic task in industrial applications. To achieve a cross-domain prognosis with incomplete target domain data, a double transferable attention mechanism was developed by Cheng et al. [165] for the fine-grained transfer.

From the literature surveyed above, the current development of RUL prediction without run-to-failure data is slower than RUL

Table 2
Solutions for RUL Prediction without Run-to-failure Data.

Application scenarios	Solutions	References
RUL prediction without run-to-failure data	Domain adversarial-based	Costa *et al.*[43], Zhou *et al.*[157], Siahpour *et al.* [158], Mao *et al.* [159], Ding *et al.* [160]
	Other TL methods	Zhang *et al.* [44], Zeng *et al.* [161], Wang *et al.* [162], Ding *et al.* [163], Wang *et al.* [164], Cheng et al. [165]

prediction with run-to-failure data because degradation knowledge is difficult to learn from limited unlabeled data in the target domain. Therefore, more effective TL techniques need to be developed for RUL prediction without run-to-failure data in the target domain.

3.4. RUL prediction with fault information

In addition to whether the target domain contains run-to-failure data in industrial scenarios, another application scenario is RUL prediction with the help of working condition information. Actually, the solutions mentioned above concentrate on improving domain adaptation performance, regardless of the difference among the bearing degradation processes of different fault types or degradation stages, which leads to a decrease in the resulting RUL prediction performance [166,201]. To make full use of such knowledge, several solutions based on auxiliary tasks have emerged in recent years and are summarized in Table 3, including fault diagnosis tasks and health condition detection tasks, which are valuable for improving the accuracy of RUL prediction.

3.4.1. RUL prediction with health state information

Recently, many researchers [45,166–171] have performed much work on improving the accuracy of RUL prediction with the help of various auxiliary tasks. From the perspective of health state detection, different degradation statuses exhibit various evolutionary processes. This leads to unreliable prediction results if the differences are ignored and an RUL model is built directly. To overcome this shortcoming, Kim et al. [167] proposed a multitask learning framework based on a health condition identification task and an RUL prediction task. In the suggested framework, a shared network was designed to capture the inter-dependencies between RUL estimation and health condition identification. The experimental results demonstrated that the proposed method achieved superior performance compared to that obtained by carrying out individual RUL prediction tasks. Other alternative methods are available for solving this problem. Zhu et al. [45] constructed a hidden Markov model to adaptively detect the fault occurrence time in each working condition. Then, a TL method combining the MMD technique and a multi-layer perceptron was presented to reduce the discrepancy between the degradation feature distributions of the source and target domains. To accurately estimate the RUL value of rolling bearings, Berghout et al. [170] presented a novel semi-supervised deep TL method for RUL prediction. Different from traditional RUL prediction, which is simply considered a traditional regression task, the proposed method addresses RUL prediction problems supporting regression, clustering, and classification aspects at the same time to achieve outstanding performance.

By introducing the information of degradation stages, the temporal features more consistent with the degradation process can be extracted; thereby, accurate RUL prediction models can be obtained instead of constructing a single model for the overall degradation process. However, the introduction of multiple regressors increases the number of parameters and model complexity. Fortunately, this problem can be improved using artificial intelligence techniques. In terms of prediction accuracy, RUL prediction solutions with health state information always provide excellent prognostic results in real industrial scenarios.

3.4.2. RUL prediction with fault type information

From the aspect of fault types, several works have investigated the relationships between different fault types and degradation processes. Xia et al. [172] developed a novel RUL prediction network based on different fault types. With the help of fault mode knowledge, the RUL prediction network can learn various degradation patterns via weakly supervised domain adaptation. Furthermore, a robust LSTM ensemble network was employed to obtain comprehensive RUL prediction results. Similarly, Huang et al. [173] utilized a multi-task learning framework to improve RUL prediction performance with the help of fault types. By combining the fault diagnosis task and RUL prediction task, a shared CNN network was applied to extract multiscale features with better trends. The experimental results demonstrated the effectiveness and robustness of the proposed method. To address the problem that the fault types of bearings are different from each other, a multi-loss transfer regression CNN containing fault classification loss, regression loss, and domain adaptation loss was constructed by Li et al. [174] to achieve accurate RUL prediction with an unlabeled target dataset. Liu et al. [175] proposed a joint-loss CNN to obtain fault diagnosis and RUL prediction information in parallel by sharing partial network parameters. In fact, some generic fault information exists in fault diagnosis and RUL prediction tasks, which can be shared to improve prediction results.

In most cases, different fault types exhibit varying degradation processes, which also provide different temporal features for RUL prediction. The above solutions can achieve satisfying prediction results if fault knowledge can be utilized rationally. In summary, TL techniques have made rapid progress in the field of intelligent RUL prediction. In this section, TL-based RUL prediction solutions are discussed in detail from the perspectives of different industrial application scenarios, which help researchers and engineers choose appropriate solutions for practical industrial applications.

Table 3
Solutions for RUL prediction with fault information.

Application scenarios	Solutions	References
RUL prediction with fault information	Health condition detection	Zhu *et al.* [45], Sun *et al.* [166], Kim *et al.* [167], Cheng *et al.* [168], Xia *et al.* [169], Berghout *et al.* [170], Berghout *et al.*[171]
	Fault diagnosis	Xia *et al.* [172], Huang *et al.* [173], Li et al. [174], Liu *et al.* [175]

4. Challenges and future trends

Existing TL methods have made important contributions to the RUL prediction task for rolling bearings under different working conditions. Nevertheless, there are still some problems that need to be further studied in future work. To inspire more technological innovation and transformative breakthroughs in RUL prediction tasks, some possible challenges and future trends are analyzed as follows.

(1) Representation ability of temporal features.

Rolling bearings mostly work in complex working conditions, such as those with high loads, high temperatures, high moisture levels, and dusty areas [70,155,176], which result in data characteristics with nonlinear, nonsmooth, non-Gaussian, and low signal-to-noise ratio expressions. However, most current studies are based on a laboratory environment, which is simpler than the real-world environment. Therefore, it is extremely important to extract high-quality temporal features with tendency and monotonicity to reduce the interference caused by environmental noise, and more robust feature extraction techniques need to be further developed to meet the actual needs of industrial applications.

(2) Interpretability of deep learning.

Although deep learning-based methods have achieved advanced results in the field of RUL prediction, the "black box" characteristics limit their applicability in industrial scenarios. At present, some studies are attempting to explore the interpretability of deep neural networks from the following aspects: feature interpretability [177–179], sample interpretability [88], and deep neural network interpretability [180,181]. Exploring the interpretability of deep learning is beneficial for engineers to make more convincing decisions, which can drive the application of deep networks in industrial scenarios.

(3) Transferability assessment of different domains.

Most TL-based RUL prediction methods directly learn and transfer knowledge from the source domain. However, it is difficult to decide what knowledge or how much knowledge is appropriate to transfer from the source domain to the target domain. An inappropriate transfer strategy can cause a negative transfer problem which in turn will impact the prediction performance. A key to avoiding negative transfer is to discover and exploit the shared underlying joint distribution between the source and target domains [182,183]. Some exploratory studies, such as related source data filtering [182], distance-based TL [184–186], and negative transfer mitigation [187–189] strategies, have attempted to address this issue. Therefore, how to quantitatively evaluate the transferability is challenging and critical, which is also a potential and valuable research direction in future studies.

(4) Stability and reliability of the data-driven model.

Model construction is an essential part of prognostics. In industrial scenarios, deciding whether a model is useful largely depends on its stability and reliability. Benefiting from the powerful feature extraction ability of artificial intelligence, data-driven methods have emerged as a research hotspot in the field of RUL prediction. However, the majority of current solutions are only focused on predicting accurate residual time values, while the uncertainty quantification problem is ignored. In industrial applications, the obtained predictive results are inevitably uncertain because the bearings are subject to random factors during the prediction process [190,191]. Therefore, it is necessary to develop new methods that can be used to describe the uncertainty bounds and confidence levels of predictive results, which is also important for RUL prediction scenarios ranging from theoretical research to practical industrial applications. Extensive research on how to effectively fuse mechanistic knowledge and improve the generalization performance of data-driven models is expected to be explored in the future.

(5) Privacy protection of industrial data.

In industrial scenarios, it is extremely important to protect data privacy, which can avoid potential conflicts of interest. Federated learning (FL), as a distributed machine learning paradigm, can effectively solve this problem. By introducing edge computing, FL allows each participant to model jointly without sharing private data, technically breaking down data silos and enabling RUL prediction collaboration. At present, FL-based studies are aimed at addressing fault diagnosis problems, and there is little work on FL-based RUL prediction [192–194]. Guo et al. [192] constructed a server and multiple edge clients to train an RUL prognostic model without sharing data, which can also reduce communication costs. Similarly, from the perspective of protecting data privacy, federated SVM and federated LSTM were developed by Bemani et al. [193] for early fault detection and RUL prediction in industrial applications. In the future, more constructive works will be investigated to speed up the process of engineering-oriented research.

(6) RUL prediction in real industrial applications.

Rolling bearings working in real industrial applications are prone to have different fault modes and degradation processes due to internal random factors and external complex environments, which results in multiple or different faults being found in the same bearing simultaneously in practical situations [69,195,196]. However, the existing RUL prognostic techniques rarely take this into account. Therefore, the ability to consider the diversity and complexities of fault modes is crucial to develop a reliable and appropriate prediction method in industrial applications.

(7) Online RUL prediction for predictive maintenance.

In most data-driven RUL prediction methods, a large amount of labeled historical data is required to train effective prognostic models [23,43,44]. However, this is challenging in real industrial applications because a faulty bearing can lead to the breakdown of other components or even the total machine. In terms of the stream data and the stochastic characteristics of its degradation process, a few statistical models combining Bayesian networks have been used in an attempt to solve online RUL prediction tasks [197–200]. The modeling strategy provides promising solutions for online RUL prediction in industrial scenarios by updating the model parameters with sequential-arrival data. At present, online RUL prediction is at the initial research stage, and it is urgent to develop more online RUL prediction methods to help engineers implement maintenance in advance.

5. Conclusion

TL is a valid and promising tool for solving rolling bearing RUL prediction problems under different working conditions. This review introduces the basic definition related to TL-based RUL prediction and summarizes the typical TL algorithms that are commonly used in RUL prediction, helping researchers quickly understand the current research progress. The state-of-the-art TL-based RUL prediction solutions are detailed and discussed from the perspectives of industrial application scenarios, specifically RUL prediction with run-to-failure data, RUL prediction without run-to-failure data, and RUL prediction with fault information. Researchers can select suitable TL methods based on specific industrial scenarios. Finally, several challenges and future study directions are analyzed. In the future, an increasing number of advanced and effective RUL prediction methods for real-world industrial applications will be studied to promote the vigorous development of RUL prediction in industry.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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